ChronoFork⏳🔀:

Immersive Learner–Multi-Agent Co-Roleplay for Divergent Historical Experiences

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https://github.com/IS492-SP26/team-project-chronofork-history-rewrite-simulation

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1. What We Needed to Validate

- What we validated:
  - Sub-task capability
  - Can they compose an experience?
- Feature priorities:
  - Checkpoint branching • Guidance • Reflection • Plausibility

2. Prompting Study Design

Typical Case → **ChatGPT**

Edge Case → **Claude**

Failure Case → **Perplexity**

Scenario

Same event + same prompts

3.1 Result: What Worked

What worked (all tools):

- Roleplay ✅ Causality ✅ Safety ✅

Cuban Missile Crisis — Decision Checkpoints (October 1962)

Checkpoint 1 — Discovery of Soviet Missiles

Date/Time: 16 Oct 1962, morning (Washington DC)
Main Actors: John F. Kennedy, McGeorge Bundy, CIA analysts, ExComm

Decision:
How should the United States respond to confirmation that the Soviet Union is installing nuclear missiles in Cuba?

Options available at the time:

- **Immediate air strike** on missile sites (preemptive attack).
- **Full invasion of Cuba** to remove missiles and Castro regime.
- **Naval blockade/quarantine** to stop further shipments and force negotiations.
- **Secret diplomatic pressure** on Moscow without public confrontation.

Canonical Historical Choice:
Kennedy forms the **Executive Committee (ExComm)** and delays action while options are debated.

Immediate Stakes & Constraints:

- Missile sites may become operational within weeks.
- Surprise attack could resemble **Pearl Harbor**, raising moral/political concerns.
- Intelligence uncertain about number and readiness of missiles.
- Risk of rapid escalation with the Soviet Union.

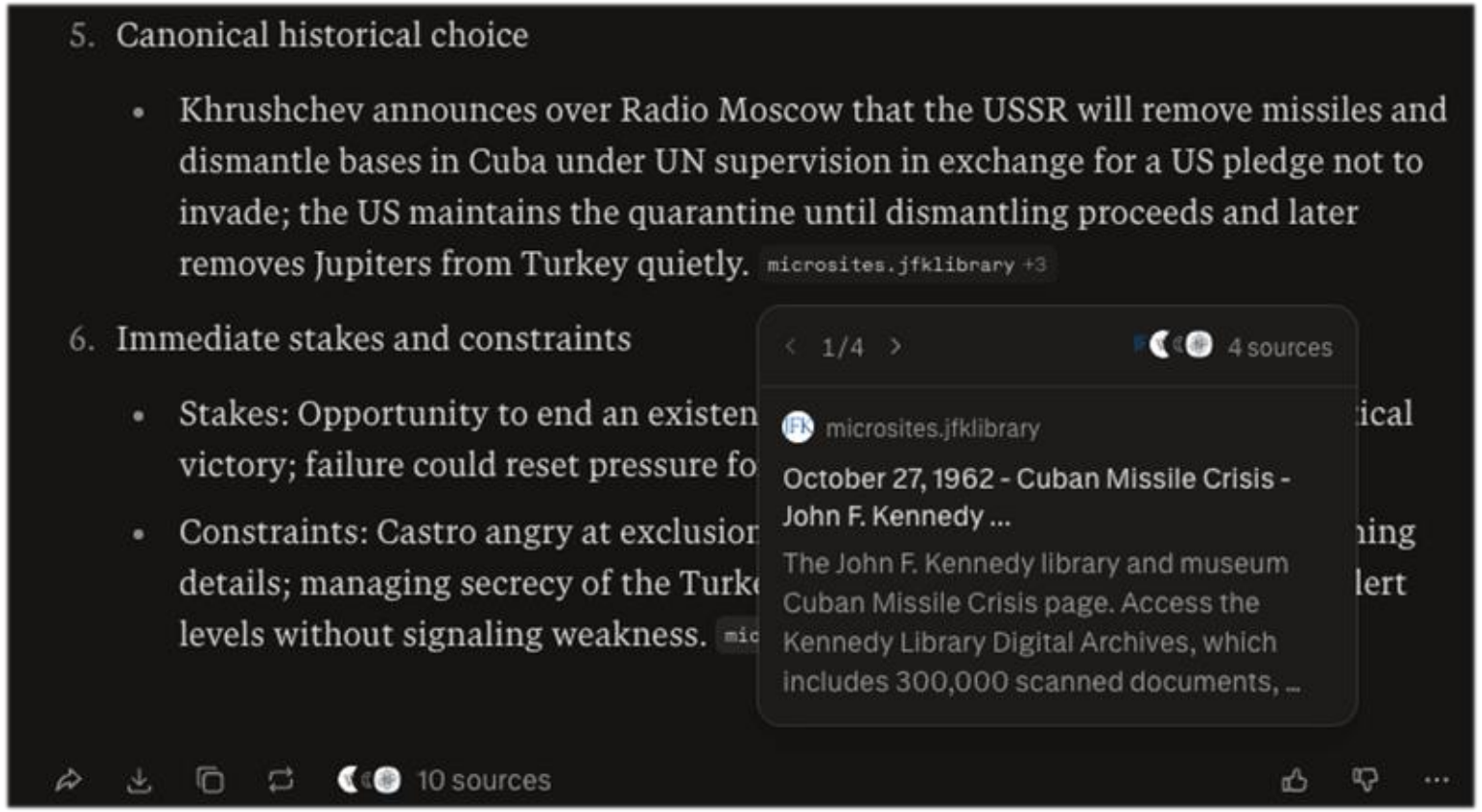
[Facilitator]
We're in the Cabinet Room on the evening of October 20, 1962, just before President Kennedy decides whether to proceed with a naval "quarantine" or a surprise air strike on the Soviet missile sites in Cuba. The ExComm has been meeting in secret for days; U-2 photos confirm Soviet medium- and intermediate-range missiles, but the sites' readiness is still uncertain. Let's listen as they focus on whether to strike or to blockade.

[JFK]
We're talking about a course of action that, once taken, we can't easily pull back from. An air strike may destroy the missiles, but it also kills a lot of Russians without warning, and that puts us very close to war they can't ignore. What I need to know is whether there's any way to protect our security without putting ourselves in the position of having fired the first shot.

3.1 Result: What Worked

Perplexity:

- Search + references ✅



3.2 Result: What Failed

What failed:

- Experience composition ❌

Make sure the explanation reflects his position rather than simply restating the U.S. perspective.

2/2

Checkpoint Context (from Moscow's perspective)
Late October 1962. The United States has announced a naval quarantine around Cuba after discovering Soviet missile deployments. In this counterfactual scenario, **President Kennedy has also publicly signaled—earlier than in reality—that the U.S. might privately accept removal of its Jupiter missiles in Turkey as part of a settlement**, while maintaining a firm public demand that Soviet missiles leave Cuba.

Below is the situation as **Nikita Khrushchev** and the Soviet leadership might reasonably interpret it.

If helpful, I can also show **how historians estimate probability changes in these branches** (e.g., likelihood of earlier settlement vs. alliance friction vs. Soviet bargaining escalation), which is useful when designing **AI-driven educational simulations.**

Not Helpful!

✅ **Good at**

Roleplay Retrieval

Causality Safety

❌ **Bad at**

Checkpoint Guidance Branch

Backtrack Engagement

4. Gap Analysis & Opportunity

Opportunity: orchestrate strong modules into a learner-centered workflow

❌ Bad at

Checkpoint **Guidance** **Branch**

Backtrack **Engagement**

→

🌟 Opportunity

Reflective report

Checkpoint DAG **Branch + Compare**

Facilitator + Tips **Plausibility rail**

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Opportunity

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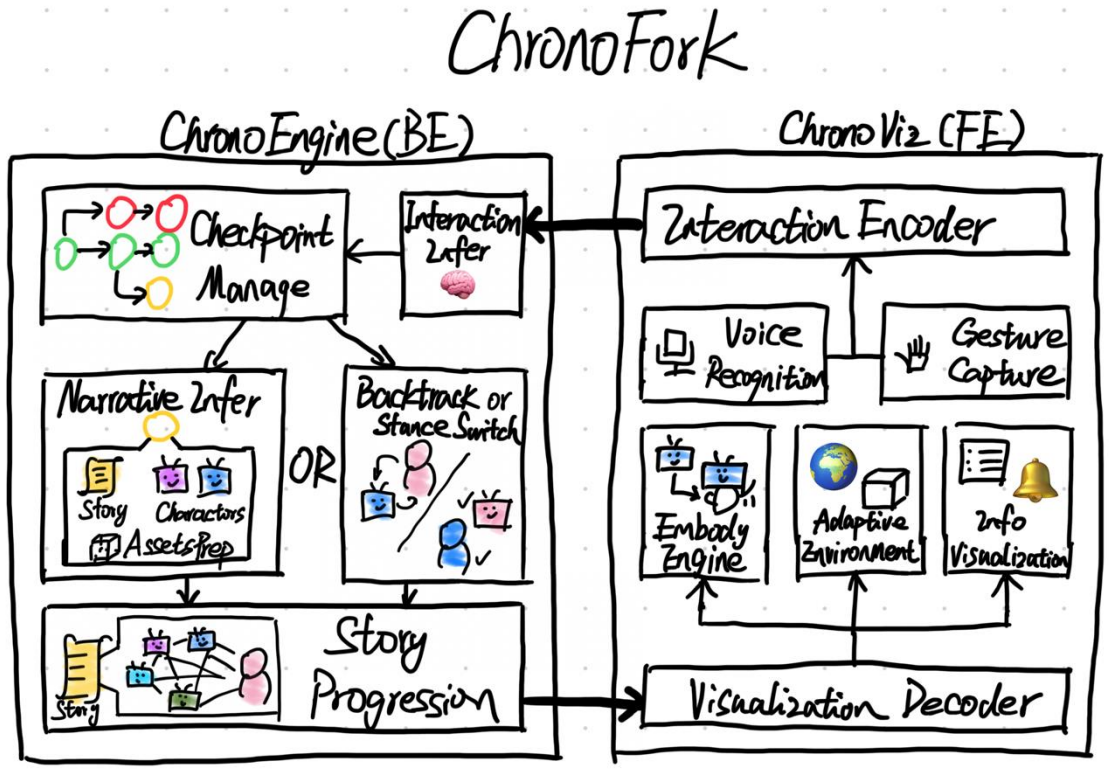
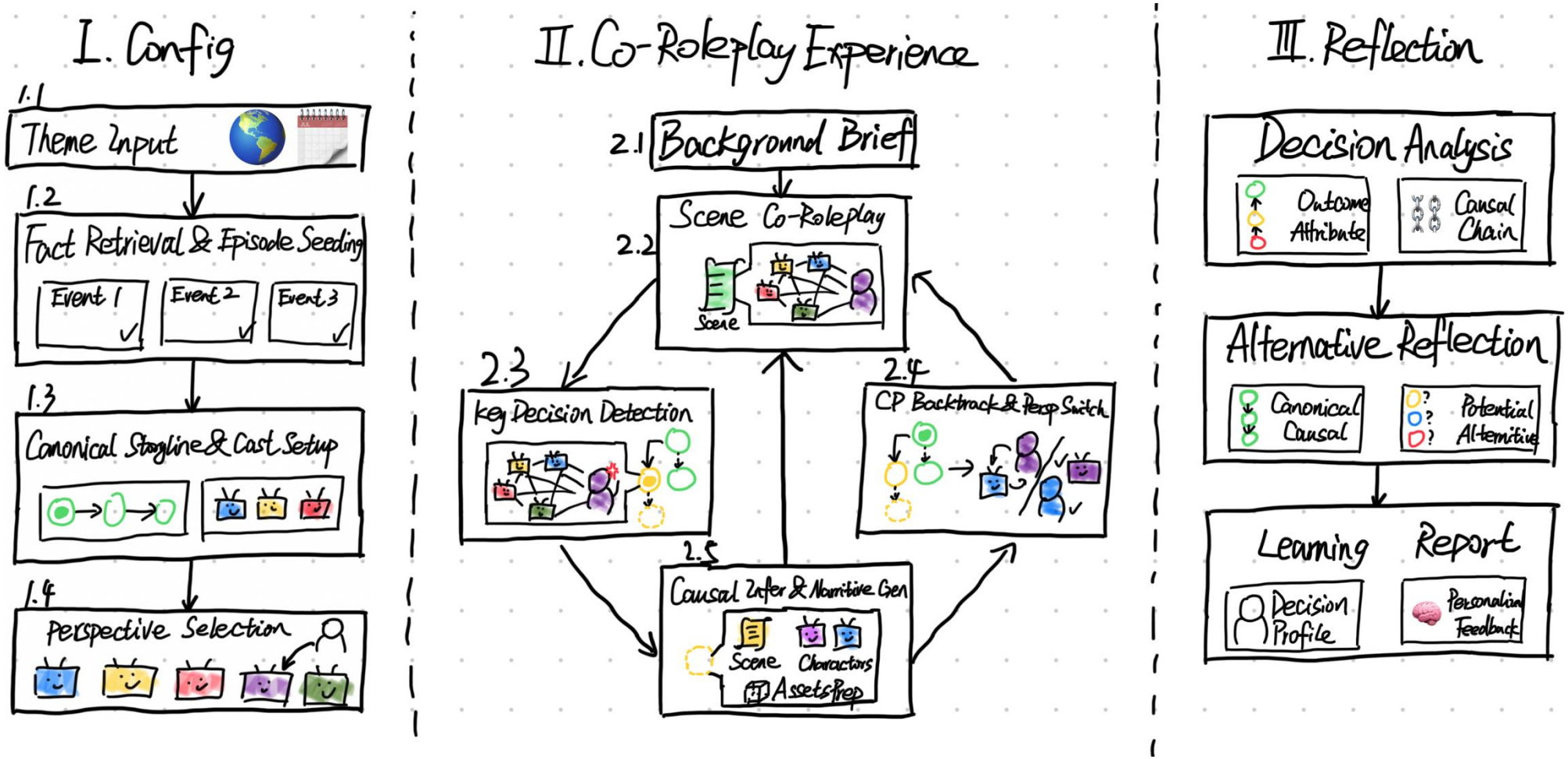


5. Refined Concept & Design Direction



6. Risks, Safeguards & Path to CP3

Risks

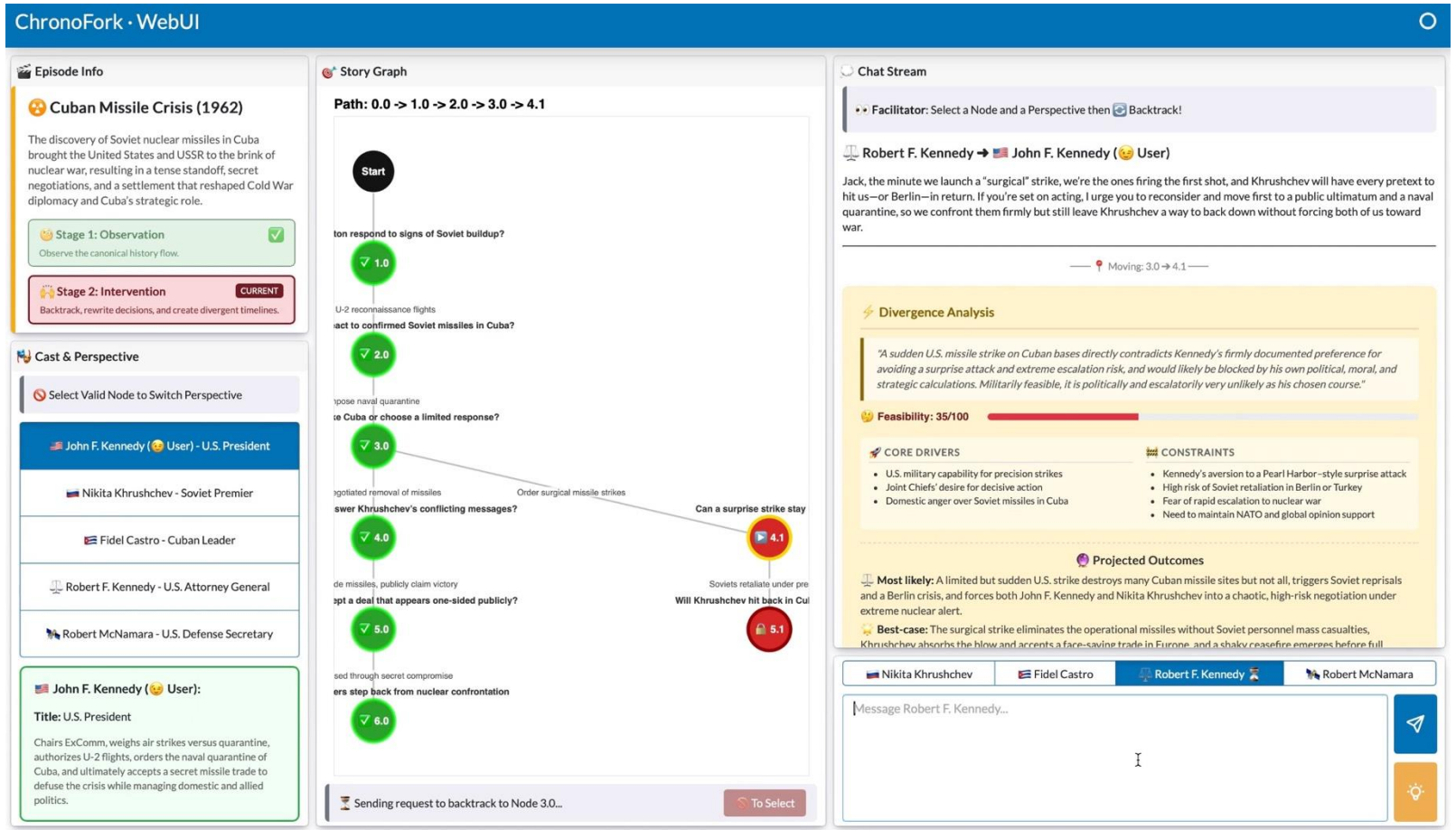
- Hallucination • Drift • Cost

Safeguards

- Grounding + plausibility rails

CP3 Build

- Stable Engine + Visualization



Thank You

Q&A

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